

Straight Edge and Compass Constructions

UMass Recreational Math Club

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You are only allowed to use the ruler to draw straight lines, not to measure things!

Beginner Constructions to try:

- Given a line, construct its perpendicular bisector.
- Given an angle, construct its bisector
- Given a line segment, construct an equilateral triangle containing that line segment as a side.
- Construct the midpoint of a given line segment

Intermediate Constructions to try:

- Given a line and a point, construct a line that point that is parallel to the given line.
- Given a line segment, divide it into 3 equal pieces.
- Given two points q and s of distance r apart and a third point p , show that you can construct a circle of radius r centered at p .
- Given a line segment of length d , construct a line segment of length $\sqrt{d}$.

Advanced Constructions to try:

- Construct a regular hexagon
- Construct a regular pentagon
- Construct a square inscribed inside a circle.